

LEVEL 2 MEDIUM QUESTIONS(4/8/26) JAVA

1.

Q1 Matrix multiplication

10 marks

Matrix multiplication

Your Code

```
1 import java.util.*;
2
3 class Main {
4     public static void main(String[] args) {
5         Scanner s = new Scanner(System.in);
6         int[][] a = new int[2][2], b = new int[2][2], c = new int[2][2];
7
8         for (int i = 0; i < 2; i++)
9             for (int j = 0; j < 2; j++)
10                 a[i][j] = s.nextInt();
11
12         for (int i = 0; i < 2; i++)
13             for (int j = 0; j < 2; j++)
14                 b[i][j] = s.nextInt();
15
16         for (int i = 0; i < 2; i++)
17             for (int j = 0; j < 2; j++)
18                 for (int k = 0; k < 2; k++)
19                     c[i][j] += a[i][k] * b[k][j];
20
21         for (int i = 0; i < 2; i++) {
22             for (int j = 0; j < 2; j++)
23                 System.out.print(c[i][j] + " ");
24             System.out.println();
25         }
26     }
27 }
```

Input

```
1 2
3 4
5 6
7 8
```

Output

```
19 22
43 50
```

2.

Q2 . Largest number

10 marks

. Largest number

Your Code

```
1 import java.util.Scanner;
2 public class Main
3 {
4     public static void main(String[] args)
5     {
6         Scanner sc = new Scanner(System.in);
7         int n = sc.nextInt();
8         int []a=new int[n];
9         for (int i =0;i<n;i++)
10        {
11            a[i]=sc.nextInt();
12        }
13        int max=a[0];
14        for(int i = 1;i<n;i++)
15        {
16            if(a[i]>max)
17            {
18                max=a[i];
19            }
20        }
21        System.out.println(max);
22    }
23 }
```

Input

4
45 85 69 1

Output

85

3.

Q3 . Smallest number

10 marks

Smallest number

Your Code

```
1 import java.util.Scanner;
2 public class Main{
3     public static void main(String[] args)
4     {
5         Scanner sc = new Scanner(System.in);
6         int n = sc.nextInt();
7         int []a=new int[n];
8         for(int i =0;i<n;i++)
9         {
10            a[i]=sc.nextInt();
11        }
12        int min=a[0];
13        for(int i=1;i<n;i++)
14        {
15            if(a[i]<min)
16            {
17                min=a[i];
18            }
19        }
20        System.out.println(min);
21    }
22 }
```

Input

3
485 652 253

Output

253

Run

Save

4.

Q4 . Nth largest number

10 marks

. Nth largest number

Your Code

```
1 import java.util.Scanner;
2 import java.util.Arrays;
3 public class Main
4 {
5     public static void main(String[] args)
6     {
7         Scanner sc = new Scanner(System.in);
8         int n=sc.nextInt();
9         int[]a=new int[n];
10        for(int i =0;i<n;i++)
11        {
12            a[i]=sc.nextInt();
13        }
14        int k=sc.nextInt();
15        Arrays.sort(a);
16        System.out.println(a[n-k]);
17    }
18 }
```

Run

Save

Input

```
3
568 236 958
2
```

Output

```
568
```

5.

Q5 Palindrome number/string

10 marks

Palindrome number/string

Your Code

```
1 import java.util.Scanner;
2 public class Main
3 {
4     public static void main(String []args)
5     {
6         Scanner sc = new Scanner(System.in);
7         String input=sc.next();
8         boolean isNumber=true;
9         for(int i=0;i<input.length();i++)
10        {
11            if(!Character.isDigit(input.charAt(i)))
12            {
13                isNumber=false;
14                break;
15            }
16        }
17        if(isNumber)
18        {
19            int n=Integer.parseInt(input);
20            int temp=n, rev=0;
21            while(n>0)
22            {
23                int rem=n%10;
24                rev=rev*10+rem;
25                n=n/10;
26            }
27            if(temp==rev)
28                System.out.println("Palindrome");
29            else
```

Input

hello

Output

Not Palindrome

```
29            else
30                System.out.println("Not Palindrome");
31        }
32        else
33        {
34            String rev="";
35            for(int i = input.length()-1;i>=0;i--)
36            {
37                rev=rev+input.charAt(i);
38            }
39            if(input.equals(rev))
40                System.out.println("Palindrome");
41            else
42                System.out.println("Not Palindrome");
43        }
44    }
45 }
```

Run

Save

6.

Q6 Removing duplicates in array

10 marks

Removing duplicates in array

Your Code

```
1 import java.util.Scanner;
2 public class Main
3 {
4     public static void main(String []args)
5     {
6         Scanner sc=new Scanner(System.in);
7         int n = sc.nextInt();
8         int[]a=new int[n];
9         for(int i=0;i<n;i++)
10        {
11            a[i]=sc.nextInt();
12        }
13        for(int i=0;i<n;i++)
14        {
15            boolean duplicate=false;
16            for(int j=0;j<i;j++)
17            {
18                if(a[i]==a[j])
19                {
20                    duplicate=true;
21                    break;
22                }
23            }
24            if(!duplicate)
25            {
26                System.out.print(a[i]+"");
27            }
28        }
29    }
30 }
```

Input

```
8
1 2 2 3 4 5 5 6
```

Output

```
123456
```